

$$m = [-1, 2]$$

$$b = [1, 1]$$

$$x = (1, 2), (4, 10)$$

$$y = (5, 6)$$

x_1	x_2	y	$\hat{y}$
1	2	5	2
4	10	6	12

$$\hat{y} = m_1 x_1 + m_2 x_2 + b = x_m + b$$

$$\hat{y} = x_m + b$$

$$\begin{bmatrix} 1 & 2 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 + 6 \\ -4 + 20 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

$$MSE = \frac{1}{n} \sum (\hat{y} - y)^2$$

$$\begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \frac{1^2 + 11^2}{2}$$

$$= \frac{122}{2}$$

$$= 61$$

$$\text{let } u = x_m + b - y$$

$$E = \frac{u^2}{n}$$

$$\frac{dE}{du} = \frac{2u}{n}$$

$$\frac{du}{dm} = X^T$$

$$\frac{dE}{dm} = \frac{2u X^T}{n} = \frac{2 X^T (\hat{y} - y)}{n}$$

$$m = \frac{2 X^T (\hat{y} - y)}{n}$$

$$= \frac{1}{n} \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} - \begin{bmatrix} 1 + 44 \\ 3 + 110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$m_{new} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - \frac{1}{100} \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$= \begin{bmatrix} -1 - \frac{45}{100} \\ 2 - \frac{113}{100} \end{bmatrix}$$

$$= \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$E = \frac{\sum (\hat{y} - y)^2}{n} = \frac{u^2}{n}$$

$$u = \hat{y} - y$$

$$\frac{dE}{du} = \frac{2u}{n}$$

$$\frac{du}{db} = (\hat{y} - y)' = (x + b - y)$$

$$= 1 \quad \frac{dE}{db} = \frac{2y}{n}$$

$$= \frac{2(\hat{y} - y)}{n}$$

$$\frac{dz}{db} = \frac{x \begin{bmatrix} 6 \\ 12 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}}{x}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= 12$$

$$b_{new} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 100 \end{bmatrix} \begin{bmatrix} 12 \\ 12 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.12 \\ 0.12 \end{bmatrix}$$

$$= \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

Iteration 2:

$$w_1 = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

$$b_1 = \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

$$\hat{y} = wx + b$$

X_1	X_2	y_0	y_1	y_2
1	3	5	6	2.04
4	10	6	17	3.78

$$X_m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} = \begin{bmatrix} -1.45 + 2.61 \\ -5.80 + 8.70 \end{bmatrix} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \frac{1}{10} \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix} \Leftrightarrow \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix} + \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

Computing error:

Let $e = \hat{y} - y$

$$e = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

MSE:

$$MSE = \frac{1}{n} \sum (\hat{y} - y)^2 = \frac{1}{n} \sum e^2$$

$$= \frac{(-2.96)^2 + (-2.22)^2}{2} = \frac{8.7616 + 4.9284}{2} = 6.845$$

$$MSE = 6.845$$

$$\frac{dE}{dw} = \frac{2}{n} \sum X^T e = \sum X^T e$$

$$dw = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix} = \begin{bmatrix} -2.96 + 4(-2.22) \\ 3(-2.96) + 10(-2.22) \end{bmatrix} = \begin{bmatrix} -2.96 + (-8.88) \\ -8.88 + (-22.2) \end{bmatrix} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

$$m_2 = m_1 - \alpha \frac{dE}{dm}$$

$$= \begin{bmatrix} -1.45 \\ 0.84 \end{bmatrix} - 0.01 \begin{bmatrix} -11.64 \\ -31.08 \end{bmatrix} = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$$

$$m_2 = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix}$$

$$\frac{dE}{db} = \frac{1}{n} \sum ((mx+b) - y)^2$$

$$\frac{dE}{db} = \frac{2}{n} \sum (mx+b) - y$$

$$= \frac{2}{7} \sum \hat{y} - y = \sum (-2.96) + (-2.22)$$

$$= \sum \hat{y} - y = -5.18$$

$$b_2 = b_1 - \alpha \frac{dE}{db}$$

$$= \begin{bmatrix} 0.88 \\ 0.88 \end{bmatrix} - 0.01 \begin{bmatrix} -5.18 \\ -5.18 \end{bmatrix} = \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

Next iteration

$$m_2 = \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} \text{ and } b_2 = \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

x_1	x_2	y	y_{pred}
1	3	5	3.1426
4	10	6	7.4134

$$y_p = mX + b$$

$$= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times (-1.3316) + 3 \times (1.1808) \\ 4 \times (-1.3316) + 10 \times (1.1808) \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3316 + 3.5424 \\ -5.3264 + 11.808 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$= \begin{bmatrix} 2.2108 \\ 6.4816 \end{bmatrix} + \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix}$$

$$y_p = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

Calculate MSE

$$\begin{aligned} \text{MSE} &= \frac{1}{n} \sum (y_p - y)^2 = \frac{1}{n} \sum e^2 \\ &= \frac{(3.1426 - 5)^2 + (7.4134 - 6)^2}{2} \\ &= \frac{3.44993476 + 1.99769956}{2} \end{aligned}$$

$$\text{MSE} = 2.72381716 \text{ in 3 decimal places}$$

$$\boxed{\text{MSE} = 2.724}$$

$$\frac{dE}{dM} = \sum \frac{2x^T (y_p - y)}{n} \text{ where } n = 2$$

$$\frac{dE}{dM} = \sum \frac{2x^T (y_p - y)}{2}$$

$$\frac{dE}{dM} = x^T (y_p - y)$$

$$y_p - y = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\boxed{y_p - y = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}}$$

$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \times (-1.8574) + 4 \times (1.4134) \\ 3 \times (-1.8574) + 10 \times (1.4134) \end{bmatrix}$$

$$= \begin{bmatrix} -1.8574 + 5.6536 \\ -5.5722 + 14.134 \end{bmatrix}$$

$$\frac{dE}{dM} = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$M_{\text{new}} = M_{\text{old}} - \alpha \frac{dE}{dM} \quad \text{where } \alpha = 0.01$$

$$= \begin{bmatrix} -1.3316 \\ 1.1808 \end{bmatrix} - 0.01 \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

$$= \begin{bmatrix} -1.3316 - 0.037962 \\ 1.1808 - 0.085618 \end{bmatrix}$$

$$= \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix}$$

$$M_{\text{new}} = \begin{bmatrix} -1.369 \\ 1.095 \end{bmatrix}$$

$$B_{\text{new}} = \frac{dE}{dB} = \frac{\sum 2(y_p - y)}{n} \quad \text{where } n = 2$$

$$\frac{dE}{dB} = \sum (y_p - y)$$

$$\frac{dE}{db} = \begin{bmatrix} 3.4426 \\ 7.4134 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$= \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$\frac{dE}{db} = -1.8574 + 1.4134$$

$$\left(\frac{dE}{db} = -0.444 \right)$$

$$b_{\text{new}} = b_{\text{old}} - \alpha \frac{dE}{db} \quad \text{where } \alpha = 0.01$$

$$= \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} - 0.01 \begin{bmatrix} -0.444 \\ -0.444 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9318 \\ 0.9318 \end{bmatrix} - \begin{bmatrix} -0.00444 \\ -0.00444 \end{bmatrix}$$

$$b_{\text{new}} = \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix}$$

So:

$$m_3 = \begin{bmatrix} -1.369 \\ 1.095 \end{bmatrix} \quad \alpha b_3 = \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix}$$

Kelvin iteration

$$m = \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix} \quad b = \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 6 \end{bmatrix} \quad x = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$$

x_1	x_2	y	y_{pred}
1	3	5	
4	10	6	

$$\begin{aligned} y_p &= X \cdot m + b \\ &= \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix} + \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix} \\ &= \begin{bmatrix} 1 \times (-1.369562) + 3 \times (1.095182) \\ 4 \times (-1.369562) + 10 \times (1.095182) \end{bmatrix} + \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix} \\ &= \begin{bmatrix} -1.369562 + 3.285546 \\ -5.478248 + 10.95182 \end{bmatrix} + \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix} \end{aligned}$$

$$z = \begin{bmatrix} 1.915984 \\ 5.473572 \end{bmatrix} + \begin{bmatrix} 0.93624 \\ 0.93624 \end{bmatrix}$$

$$y_p = \begin{bmatrix} 2.852224 \\ 6.409812 \end{bmatrix}$$

$$MSE = \frac{\sum (y_p - y)^2}{n} = \frac{1}{n} \sum e^2$$

$$= \frac{(2.852224 - 5)^2 + (6.409812 - 6)^2}{2}$$

$$= \frac{-2.147776^2 + 0.409812^2}{2}$$

$$= \frac{4.612907 + 0.167946}{2}$$

$$MSE = 2.390427$$

$$\frac{dE}{dm} = X^T (y_p - y) \quad \left| \begin{array}{l} e/y_p - y = \begin{bmatrix} -2.147776 \\ 0.409812 \end{bmatrix} \end{array} \right.$$



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$$= \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.147776 \\ 0.409812 \end{bmatrix}$$

$$= \begin{bmatrix} 1(-2.147776) + 4(0.409812) \\ 3(-2.147776) + 10(0.409812) \end{bmatrix}$$

$$\frac{dE}{dm} = \begin{bmatrix} -0.508528 \\ -2.345208 \end{bmatrix}$$

$$m_{new} = m_{old} - \alpha \frac{dE}{dm} \quad \text{where } \alpha = 0.01$$

$$= \begin{bmatrix} -1.369562 \\ 1.095182 \end{bmatrix} - \begin{bmatrix} -0.508528 \\ -2.345208 \end{bmatrix} \times 0.01$$

$$= \begin{bmatrix} -2.861034 \\ 3.440390 \end{bmatrix} \begin{bmatrix} -1.36447672 \\ 1.11863408 \end{bmatrix}$$

$$\frac{dE}{db} = \sum e$$

$$e = \begin{bmatrix} -2.147776 \\ 0.409812 \end{bmatrix}$$

$$\frac{dE}{db} = -2.147776 + 0.409812$$

$$= -1.737964$$



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$$m_{new} = m_{old} - \alpha \frac{dE}{db}$$

$$= \begin{bmatrix} 0.93624 \\ 0.9364 \end{bmatrix} - 0.01 \begin{bmatrix} -1.737964 \\ +1.737964 \end{bmatrix}$$

$$= \begin{bmatrix} 0.95361964 \\ 0.95361964 \end{bmatrix} \quad \text{to 4 decimal places}$$

$$b_{new} = \begin{bmatrix} 0.9536 \\ 0.9536 \end{bmatrix}$$

So

$$m_{new} = \begin{bmatrix} -1.3645 \\ 1.1186 \end{bmatrix} \quad b_{new} = \begin{bmatrix} 0.9536 \\ 0.9536 \end{bmatrix}$$